

Solution Requirements

Date	5 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	No user login is required. The flood prediction application can be accessed directly without requiring users to create an account or log in.	Low
2	Authorization levels	Single user access. The system provides the same functionality to all users and does not support multiple user roles or permission-based access.	Low
3	External interfaces	The Flask application communicates with the trained machine learning model	High

4	Transactions processing	Users provide weather-related inputs, which are verified by the system before being processed by the machine learning model to generate a flood risk prediction.	High
5	Reporting	The application presents the prediction outcome by indicating whether the selected conditions represent a Flood Risk or No Flood Risk .	Medium
6	Business rules, etc.	All weather parameters must be entered as valid numerical values. The prediction is generated using the trained machine learning model and historical weather data.	High
7	Compliance to laws or regulations	The application does not collect or retain personal user information and follows standard data privacy and security practices.	Medium
8	Other	Users can submit new weather information and perform multiple flood predictions without reloading or restarting the application.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application should generate flood prediction results promptly after receiving the user's weather inputs.	Prediction completed within 2 seconds
2	Scalability	The system should efficiently handle multiple users requesting flood predictions at the same time after deployment.	Supports more than 100 simultaneous users
3	Security & Data Privacy	The application must validate all user inputs and avoid storing any personal or sensitive information.	Robust input validation with zero permanent data storage
4	Reliability & Availability	The flood prediction service should remain operational with very little interruption or downtime.	Minimum 99% system availability
5	Usability & Accessibility	The interface should be simple, responsive and easy for users to understand.	Easy navigation with clear input fields and result display
6	Other	The system should be designed to support future enhancements, including real-time weather integration and upgraded machine learning models.	Modular Flask architecture with reusable code